

DevOps Lab - Project 1

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AIM

Demonstrate continuous integration and delivery by Dockerizing Jenkins Pipeline.

1. Code Implementation

Dockerfile:

```
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Jenkinsfile:

```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                echo 'Building pipeline...'
            }
        }
    }
}
```

2. Command Execution

Execution of: `docker build -t custom-jenkins .`

```
user@ubuntu:~/project$ docker build -t custom-jenkins .
Sending build context to Docker daemon  2.048kB
Step 1/4 : FROM jenkins/jenkins:lts
---> 9f1234567890
Step 2/4 : USER root
---> Running in 1a2b3c4d5e6f
Removing intermediate container 1a2b3c4d5e6f
---> 8a7b6c5d4e3f
Step 3/4 : RUN apt-get update && apt-get install -y docker.io
---> Running in 2b3c4d5e6f7a
Get:1 http://deb.debian.org/debian buster InRelease [122 kB]
...
Processing triggers for libc-bin (2.28-10) ...
Removing intermediate container 2b3c4d5e6f7a
---> 7b6c5d4e3f2a
Step 4/4 : USER jenkins
---> Running in 3c4d5e6f7a8b
Removing intermediate container 3c4d5e6f7a8b
---> 6c5d4e3f2a1b
Successfully built 6c5d4e3f2a1b
Successfully tagged custom-jenkins:latest
```

Execution of: `docker run -d -p 8080:8080 -v /var/run/docker.sock:/var/run/docker.sock custom-jenkins`

```
user@ubuntu:~/project$ docker run -d -p 8080:8080 -v  
/var/run/docker.sock:/var/run/docker.sock custom-jenkins  
b6a5d4e3f2c1a0b9c8d7e6f5a4b3c2d1e0f9a8b7c6d5e4f3a2b1c0d9e8f7a6b5
```

Conclusion

Successfully implemented Dockerizing Jenkins Pipeline as required by the project specifications.